

Welcome to

Become a Python Developer

Course Instructor: Md. Azizul Hakim

Python divides the operators in following groups

- Arithmetic operators // +,-
- Assignment operators, =
- Comparison operators
- Logical operators
- Membership operators
- Bitwise operators

Some examples of Arithmetic operators

Operator	Example
+	$x+y$
-	$x-y$
*	$x*y$
/	x/y
%	$x\%y$
**	$x**y$
//	$x//y$

Some examples of assignment operators

Operator	Example
=	X=10
+=	x+=y
-=	x-=y
=	x=y
/=	x/=y
%=	x%=y
=	x=y
//=	x//=y

Some examples of Comparison operators

Operator	Example
==	X==5
!=	X!=3
>	X>2
<	X<2
>=	X>=5
<=	X<=2

Examples of Logical operators

Operator	Example
and	$X > 5$ and $Y > 5$
or	$X > 5$ or $Y > 5$
not	not ($X > 2$ and $X < -3$)

Examples of Membership operators

Operator	Example
in	X in Y
not in	X not in Y

Examples of Bitwise operators

Operator	Example
&	X & Y
	X Y
^	X ^ Y
~	X ~ Y
<<	X << 2
>>	X >> 2

if syntax

if condition:

body of if statement

Example

```
number = 3
```

```
# check if number is greater than 0
```

```
if number > 0:
```

```
    print('Number is positive.')
```

```
    print('lets learn conditional')
```

if syntax

if condition:

 # block of code if condition is True

else:

 # block of code if condition is False

Example

```
number = int(input('enter a number: '))
```

```
if number > 0:
```

```
    print('Positive number')
```

```
else:
```

```
    print('Negative number')
```

```
print('This statement is always executed')
```

Conditional Statement

if...elif.else

syntax

```
if condition1:  
    # code block 1  
  
elif condition2:  
    # code block 2  
  
else:  
    # code block 3
```

Example

```
number = 0  
  
if number > 0:  
    print("Positive number")  
  
elif number == 0:  
    print('Zero')  
else:  
    print('Negative number')  
  
print('This statement is always executed')
```

syntax

```
# outer if statement
if condition1:
    # statement(s)

# inner if statement
if condition2:
    # statement(s)
```

Example

```
number = 5
# outer if statement
if (number >= 0):
    # inner if statement
    if number == 0:
        print('Number is 0')
    # inner else statement
    else:
        print('Number is positive')
# outer else statement
else:
    print('Number is negative')
# Output: Number is positive
```

loops are used to repeat a block of code

syntax

```
for val in sequence:  
    # statement(s)
```

Example

```
for x in 'Python':  
    print(x)
```

```
#another loop  
for x in range(6):  
    print(x)  
for x in range(2, 6):  
    print(x)  
for x in range(2, 30, 3):  
    print(x)
```

loops are used to repeat a block of code

syntax

while condition:

 # body of while loop

Example

program to display numbers from 1 to 5

initialize the variable

i = 1

n = 5

while loop from i = 1 to 5

while i <= n:

 print(i)

 i = i + 1

Break is used to terminate the loop immediately when it is encountered

continue is used to skip the current iteration of the loop

Example

```
#break
```

```
for i in range(5):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

```
#continue
```

```
for i in range(5):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```